



Come to HUFS Meet the World

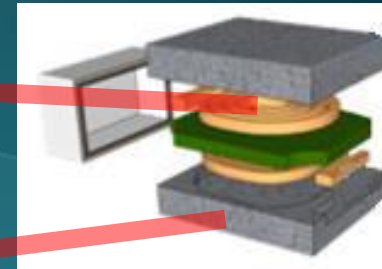


A Study on the Magnetic Characteristics of Power Inductor with Metal Nanopowder

◆ Power Inductor

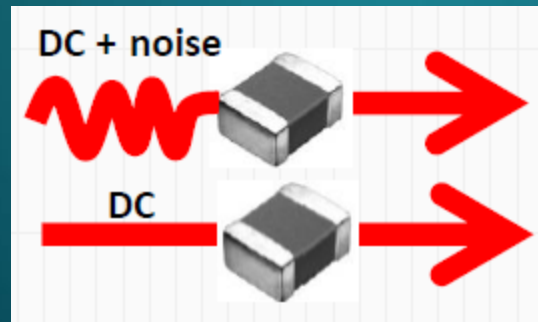
1) Configuration

- Current-flow coil
- Low loss(R_{DC}) demand
- Magnetic body material surrounding coil
- High permeability, high M_s , low loss



2) Usage

Inductor requiring a high rated current at the power stage



Current Smoothing

- **Blocking** : Noise passing obstruct
- **Passing** : Rectified high capacity direct current flow smoothly

◆ Power Inductor Development Trend

1) Minimization

: High inductance with small size

$$L = \frac{\mu_0 \mu_r N^2 A_e}{l_e}$$

→ High permeability(μ_r)

2) High Currentization

: High rated current (DC-bias characteristics)

→ DC-bias characteristics are proportional to M_s

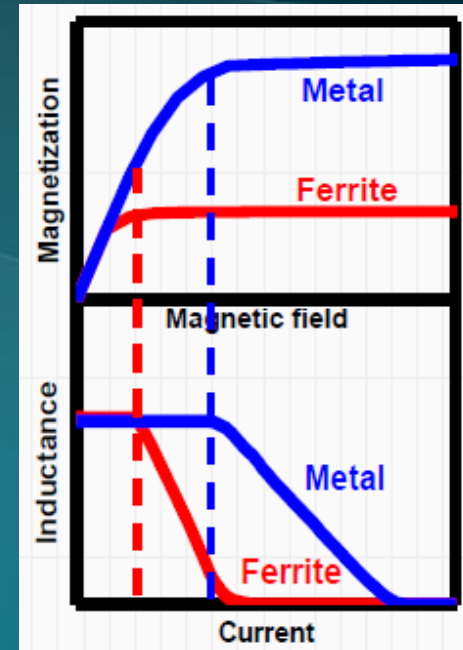
3) High Efficiency

: Core loss ↓ → $P_{CV} = P_h + P_e$

Hysteresis loss - H_c ↓

$$P_h = KhBmf$$

Eddy current loss - electrical resistivity ↑ $P_e = t \frac{B_m^2 f^2 d^2}{\rho}$



◆ Materials

1) Crude Powder : Fe-6.5wt%Si crystal powder

(about 190 emu/g, High M_s)

: In the past, ferrite. Nowadays, metal powder

(70 emu/g) (about 130 emu/g)

- Material with high M_s is used to improve DC-bias characteristics

Ollendorf's eq:
$$\mu_r = \frac{\eta \mu_0 (\mu_m - \mu_0)}{N(1-\eta)(\mu_m - \mu_0) + \mu_0}$$

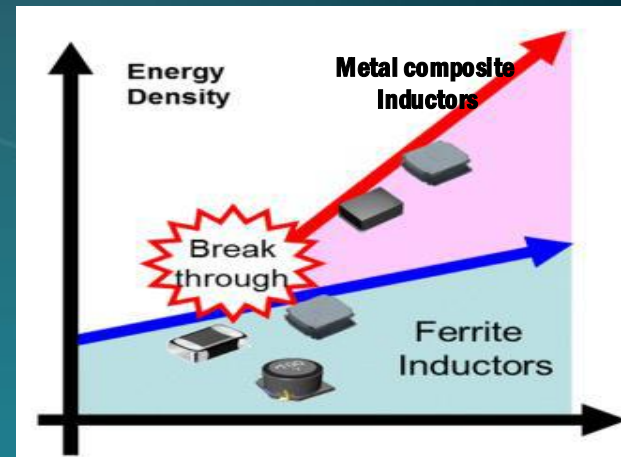
* Improvement of permeability:

- (1) Improvement of packing fraction of metal composite
- (2) Reducing N
- (3) Improving bulk permeability of metal powder

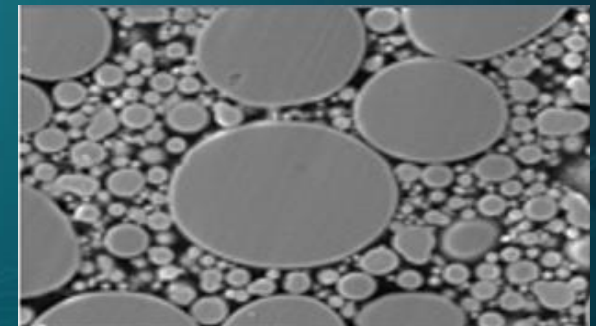
2) Small Powder : Fe nanopowder

: Now using small Fe powder with micro grain size

- Nanopowder is applied as a future power inductor material by increasing the packing fraction of metal composite by increasing the permeability



η : packing fraction
 N : demagnetizing field coefficient
 ($0 \leq N \leq 1$)
 μ_m : bulk permeability
 μ_0 : vacuum permeability



◆ Equipment



<PWE>
Nanoparticle fabrication



<VSM>
Magnetization measurement



<Impedance Analyzer>
Permeability measurement



<B-H Analyzer>
Core loss measurement



<SEM & TEM>
Powder shape and microstructure,
section of core observation



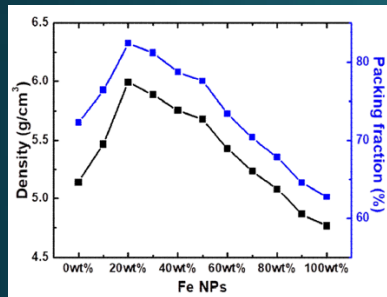
<PSA>
Particle Size Analyzer

◆ Core Characteristics

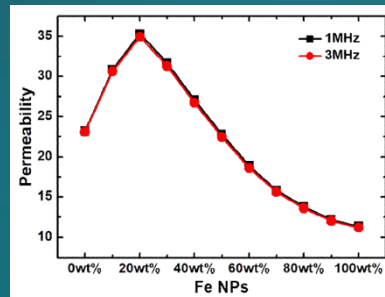
: Packing fraction of crude powder and small powder

Maximum packing fraction at 8:2 – 82.4%, permeability 35.3

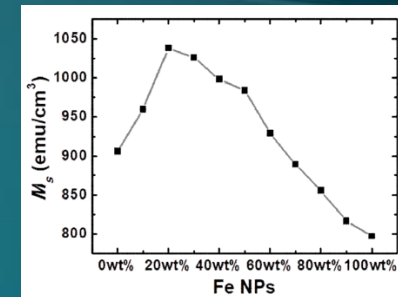
<Core density and packing fraction>



<Core permeability>

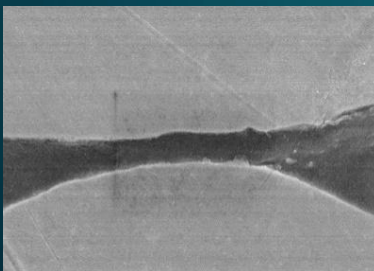


<Core magnetization>

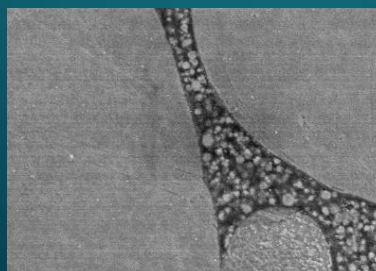


<Core Section Microstructure>

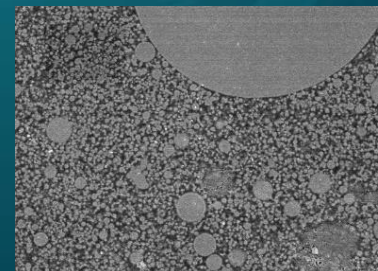
<0wt%>



<20wt%>



<50wt%>

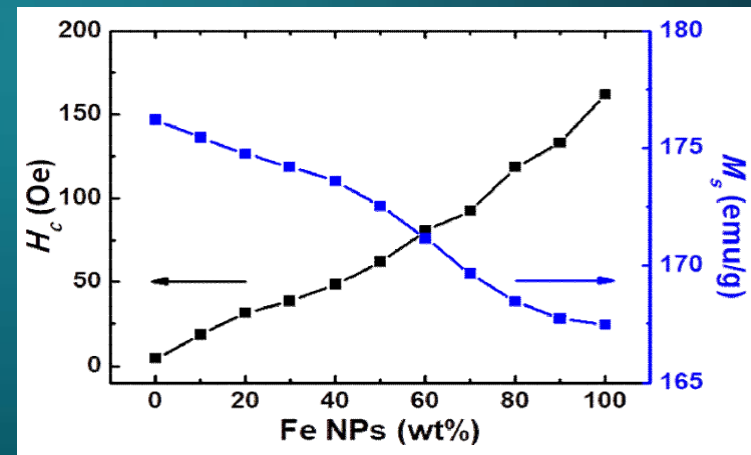
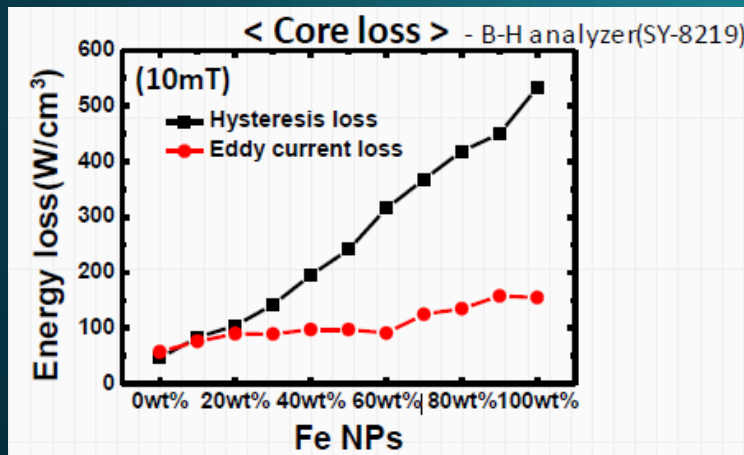


◆ Core loss

Core loss = Hysteresis loss + Eddy current loss

$$P_{CV} = Kh(B_m)^k f + Ke(B_m f)^2$$

(k : Steinmetz coefficient)



◆ A Study on the Metallic Magnetic Powder for Power Inductor

- 1) Fabricate 100nm nano-metallic powder using PWE, crude powder Fe-6.5wt%Si using gas atomizer.
 - 2) A study on the increase of permeability through the increase of packing fraction of metal magnetic compound
 - 80wt% Fe-6.5Si + 20wt% Fe nanopowder with maximum permeability of 35.3 and maximum packing fraction of 82.4%
 - Metallic magnetic complex maximum DC-bias characteristics can be implemented (High $M_s \sim 1038 \text{ emu/cm}^3$)
 - 3) Decrease in core loss when applying nano powder compared to small powder in the diameter of several μm ($650 \text{ mW/cm}^3 \rightarrow 200 \text{ mW/cm}^3$)
- ※ Power inductor characteristics can be improved by adding additional metal nanoparticles.
- Minimization of power inductor, improvement of DC-bias characteristic(High M_s), decrease of core-loss(improvement of efficiency)

**Thank you for your
concentration!**